

19-5-26

Linear Regression

Load (N)

Extension

5

1.2

10

2.5

15

3.7

20

4.9

25

6.1

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \quad (y = mx + c)$$

$$c = \frac{\sum y - m \sum x}{n}$$

Table (Calculation purpose)

x	y	x ²	xy
5	1.2	25	6
10	2.5	100	25
15	3.7	225	55.5
20	4.9	400	98
25	6.1	625	152.5

$$\sum x = 75$$

$$\sum x^2 = 1375$$

$$\sum xy = 337$$

$$\sum y^2 \Rightarrow \sum y = 18.4$$

$n \Rightarrow$ no. of observations = 5

$$m = \frac{5(\sum xy) - (\sum x)(\sum y)}{5(\sum x^2) - (\sum x)^2}$$

$$= \frac{5(337) - (75)(18.4)}{5(1375) - (75)^2}$$
$$= \frac{1685 - 1380}{6875 - 5625} = \frac{305}{1250} \approx 0.244$$

$$c = \frac{\sum y}{n} - \frac{m \sum x}{n} = \frac{18.4}{5} - \frac{0.244(75)}{5}$$

$$= \frac{18.4 - 18.3}{5} = \frac{0.1}{5} = \frac{1}{50} = \frac{1}{5} \times 10^{-1}$$

$$\Rightarrow c = \underline{\underline{0.02}}$$

Regression formula $\rightarrow y = mx + c$

$$\Rightarrow \{ y = 0.244x + 0.02 \}$$